$$C_{u\phi_2}^{(22)\,i1\_i2\_} \to \frac{1}{16\,\pi^2}\Big(\frac{1}{3} m_2 \tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] - \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] +$$
$$\frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{2} \tilde{\mu} g_2^2 y_e^{pr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \frac{1}{4} \tilde{\mu} g_1^2 y_e^{pr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] -$$
$$\frac{1}{8} \tilde{\mu} y_e^{pr} y_u^{i1i2} \overline{a_e}^{pr} (3 g_1^2 + 7 g_2^2) \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{6} \tilde{\mu} y_e^{pr} y_u^{i1i2} \overline{a_e}^{pr} (3 g_1^2 + 2 g_2^2)$$
$$\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{2} \tilde{\mu} y_d^{pr} y_u^{i1i2} \overline{a_d}^{pr} (2 g_1^2 + 3 g_2^2) \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] -$$
$$\frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \frac{21}{8} \tilde{\mu} y_d^{pr} y_u^{i1i2} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$
$$\frac{1}{2} \tilde{\mu} y_d^{pr} y_u^{i1i2} \overline{a_d}^{pr} (3 g_1^2 + 2 g_2^2) \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$
$$\frac{1}{8} \tilde{\mu} y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} (g_1^2 - g_2^2) \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \frac{3}{4} \tilde{\mu} g_2^2 y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$$
$$\frac{1}{4} \tilde{\mu} g_2^2 y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \frac{3}{4} \tilde{\mu} g_2^2 y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$$
$$\frac{1}{4} \tilde{\mu} y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} (g_1^2 - 2 g_2^2) \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$$
$$\frac{1}{8} \tilde{\mu} y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} (-g_1^2 + g_2^2) \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] -$$
$$\frac{3}{4} \tilde{\mu} y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr} (g_1^2 + 2 g_2^2) \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{2} \tilde{\mu} y_u^{pr} y_u^{i1i2} \overline{a_u}^{pr}$$
$$(3 g_1^2 + 2 g_2^2) \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] -$$
$$\frac{1}{2} m_1 \tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (6 g_1^2 + 5 g_2^2) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] +$$
$$\frac{1}{2} m_1 \tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (3 g_1^2 + 2 g_2^2) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] +$$
$$\frac{1}{12} m_2 \tilde{\mu} g_2^2 y_u^{i1i2} (9 g_1^2 + 5 g_2^2) \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] - \frac{7}{3} m_2 \tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] -$$
$$\frac{1}{8} m_2 \tilde{\mu} g_2^2 y_u^{i1i2} (18 g_1^2 + 11 g_2^2) \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] +$$
$$2 m_2 \tilde{\mu} g_2^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{2} m_2 \tilde{\mu} g_2^2 y_u^{i1i2} (3 g_1^2 + 2 g_2^2) \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] -$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}] -$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{u}}^{i2}, m_{\tilde{q}}^{i1}] +$$
$$\frac{1}{48} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (-7 g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i1}] +$$
$$\frac{1}{12} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (7 g_1^2 - g_2^2) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{u}}^{i2}] -$$
$$\frac{1}{4} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (7 m_1 + 5 m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] +$$
$$\tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{3}{16} m_2 \tilde{\mu} g_2^2 y_u^{i1i2} (g_1^2 - 7 g_2^2)$$
$$\mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i1}] - \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}] -$$
$$\frac{1}{6} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde{u}}^{i2}, m_{\tilde{q}}^{i1}] + \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr}$$
$$\mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}] + \frac{1}{24} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (7 g_1^2 + 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] -$$
$$\frac{1}{8} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, \tilde{\mu}] - \frac{1}{16} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (g_1^2 + g_2^2)$$
$$\mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{p}] - \frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] +$$
$$\frac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}] - \frac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr}$$
$$\mathrm{LF}_{2,2,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \frac{1}{2} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] +$$
$$\frac{1}{2} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}] + \frac{1}{4} \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr}$$
$$\mathrm{LF}_{2,2,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}] - \tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] +$$
$$\tilde{\mu} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr}$$
$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] + \frac{3}{2} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}] +$$
$$\frac{1}{24} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (5 g_1^2 + 9 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, \tilde{\mu}] -$$
$$\frac{1}{8} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, \tilde{\mu}] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}] -$$
$$\frac{1}{16} \tilde{\mu} y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} (g_1^2 + g_2^2) \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, \tilde{\mu}, m_{\tilde{d}}^{r}] -$$
$$3 \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$$
$$3 \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{4} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{8} \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] +$$
$$\frac{1}{108} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (5 g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{i2}] -$$
$$\frac{1}{36} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_1, m_{\tilde{u}}^{i2}] +$$
$$\frac{1}{72} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_1, \tilde{\mu}] -$$
$$\frac{1}{8} m_2 \tilde{\mu} g_2^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_2, \tilde{\mu}] +$$
$$\frac{1}{9} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (5 g_1^2 - 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{u}}^{i2}] -$$
$$\frac{1}{3} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{i1}, m_3, m_{\tilde{u}}^{i2}] +$$
$$\frac{1}{108} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (-5 g_1^2 + 3 g_2^2) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{i2}, m_1, m_{\tilde{q}}^{i1}] -$$
$$\frac{1}{36} m_1 \tilde{\mu} g_1^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{i2}, m_1, m_{\tilde{q}}^{i1}] +$$
$$\frac{2}{9} m_1 \tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{i2}, m_1, \tilde{\mu}] + \frac{1}{9} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (-5 g_1^2 + 3 g_2^2)$$
$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{i2}, m_3, m_{\tilde{q}}^{i1}] - \frac{1}{3} m_3 \tilde{\mu} g_3^2 y_u^{i1i2} (g_1^2 + g_2^2) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{i2}, m_3, m_{\tilde{q}}^{i1}] +$$
$$\frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_1, m_2] - \frac{1}{8} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (9 m_1 + 5 m_2) \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] +$$
$$\frac{3}{4} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] + \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2)$$
$$\mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \frac{2}{9} m_1 \tilde{\mu} g_1^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}] +$$
$$m_2 \tilde{\mu} g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] - \tilde{\mu} \overline{y_d}^{st} y_d^{sr} y_d^{i1t} y_u^{pi2} \overline{a_d}^{pr}$$
$$\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, \tilde{\mu}] - \tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, \tilde{\mu}] -$$
$$\frac{3}{4} \tilde{\mu} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +$$
$$\frac{3}{4} \tilde{\mu} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] -$$
$$\frac{1}{4} \tilde{\mu} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{4} \tilde{\mu} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] -$$
$$\frac{1}{2} \tilde{\mu} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +$$
$$\frac{1}{2} \tilde{\mu} y_e^{sr} y_u^{i1i2} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}] +$$
$$\frac{3}{4} \tilde{\mu} y_u^{pt} y_u^{i1i2} \overline{a_d}^{pr} (a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] -$$
$$\frac{3}{4} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{4} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] -$$
$$\frac{3}{8} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{8} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{2} \tilde{\mu} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] +$$
$$\frac{3}{2} \tilde{\mu} y_d^{sr} y_u^{i1i2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] -$$
$$\frac{3}{4} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{4} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] +$$
$$\frac{9}{8} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{8} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$$
$$\frac{3}{2} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{2} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] +$$
$$\frac{3}{4} \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] +$$
$$\frac{1}{6} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}^{i1}] +$$
$$\frac{1}{3} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{u}}^{i2}] +$$
$$\frac{1}{2} m_1 g_1^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] -$$
$$\tilde{\mu} g_2^2 y_d^{i1r} y_u^{pi2} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] +$$
$$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] +$$
$$\frac{1}{3} \tilde{\mu} g_1^2 g_2^2 y_u^{i1i2} (m_1 + m_2) \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde{q}}^{i1}, m_{\tilde{u}}^{i2}, \tilde{\mu}] +$$
$$\frac{2}{3} m_1 g_1^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$$
$$\frac{1}{6} m_1 g_1^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$$
$$\frac{2}{9} m_1 g_1^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}] +$$
$$m_2 \tilde{\mu} g_2^2 y_u^{pi2} \overline{a_d}^{pr} a_d^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, \tilde{\mu}] +$$
$$\frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, \tilde{\mu}] -$$
$$\frac{8}{3} m_3 g_3^2 \tilde{\mu}^3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_3, m_{\tilde{q}}^{i1}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, m_{\tilde{u}}^{r}] -$$
$$\tilde{\mu} y_d^{i1r} y_u^{si2} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, \tilde{\mu}]\Big)$$